Chapter 1

(AST305) Lifetime Data Analysis I

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Lecture Outline

1. 1. Basic Concepts and Models
   - 1.1 Introduction
   - 1.2 Lifetime Distributions
   - 1.3 Some Important Failure Time Models
   - 1.4 Regression Models

Section 1

1. Basic Concepts and Models

Subsection 1

1.1 Introduction

Lifetime Data

- **Lifetime data** have important use in many research areas, including health sciences, engineering, social sciences, etc.
- Applications range from investigating the **durability of manufactured products** to studying **human diseases and their treatments**.
- Lifetime data are also known as "**survival time data**" or "**failure time data**"

Example 1.1.1 – Engineering

- Manufactured items (mechanical or electronic) are often tested for durability.
- Items are operated under controlled conditions and observed until they fail.
- Here, lifetimes are referred to as **"failure times."**

Example 1.1.2 – Social Sciences

- Demographers study the duration of human life "states."
- Example: The **duration of marriage** formed in 1980 in a country.
- Marriage ends with **annulment, divorce, or death.**

Example 1.1.3 – Medical Studies

- For patients with potentially fatal diseases, the key outcome is **survival time**, usually measured from diagnosis or treatment.
- Treatments are often compared using the **distribution of survival times.**

Example 1.1.4 – Laboratory Experiments

- In carcinogenic studies, laboratory animals are exposed to substances and followed until tumors appear.
- The primary outcome is **time to tumor appearance**.

Time Origin and Time Scale

Time Origin

- The **zero point** from which survival time is measured for each subject.
- Marks the **start of follow-up**.
- **Examples:** In a medical study, time origin can be the *date of diagnosis*; in a clinical trial, the *date of randomization*; and in an employment study, the *date of recruitment*, all marking when the follow-up clock starts.
- **Key Points**
  - **Must be clearly defined**
  - **Can differ between individuals** (rolling recruitment)

Time Origin and Time Scale

Time Scale

- The **metric used to measure time** from the time origin.
- Determines the unit and meaning of "time" in the model.
- **Common Time Scales:**
  - *Time since study entry* (follow-up duration from study start, in hours, minutes, seconds etc.)
  - *Attained age* (participant's actual age at each point in follow-up).
- Time need not always be chronological:
  - **Miles driven** for a vehicle until breakdown.
  - **Number of pages printed** for a printer or photocopier.

Censoring

- In practice, full observation of lifetimes may not be feasible.
- If we only know a subject's lifetime **exceeds a certain value**, it is a **censored observation**.
- **Example:** A life test stops after 28 days. If an item has not failed, its lifetime is known only as **"greater than 28 days."**
- **Types of censoring:** right, left, and interval censoring (details in next chapter).

Censoring

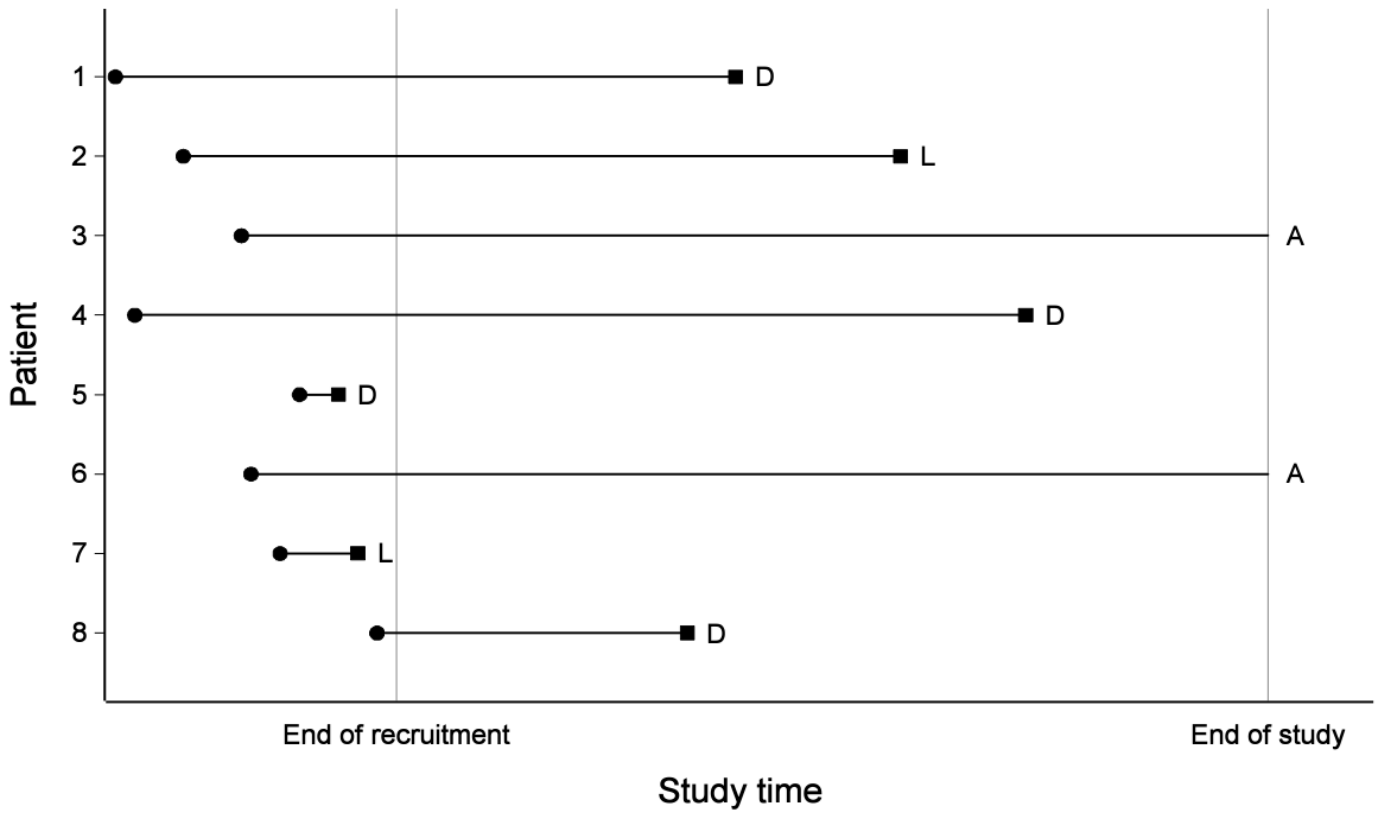


Figure 1: Lifetime of eight subjects in a survival study. D=Death, L=Lost, A=Alive

Example 1.1.5 – Electrical Insulating Fluid

- Nelson (1972) studied electrical insulating fluid under constant voltage stress.
- Failure time (time to breakdown) was recorded.
- Specimens were tested at different voltages (26–38 kV).

Table 1.1. Times to Breakdown (in minutes) at Each of Seven Voltage Levels

Voltage Level (kV)	n_i	Breakdown Times
26	3	5.79, 1579.52, 2323.7
28	5	68.85, 426.07, 110.29, 108.29, 1067.6
30	11	17.05, 22.66, 21.02, 175.88, 139.07, 144.12, 20.46, 43.40, 194.90, 47.30, 7.74
32	15	0.40, 82.85, 9.88, 89.29, 215.10, 2.75, 0.79, 15.93, 3.91, 0.27, 0.69, 100.58, 27.80, 13.95, 53.24
34	19	0.96, 4.15, 0.19, 0.78, 8.01, 31.75, 7.35, 6.50, 8.27, 33.91, 32.52, 3.16, 4.85, 2.78, 4.67, 1.31, 12.06, 36.71, 72.89
36	15	1.97, 0.59, 2.58, 1.69, 2.71, 25.50, 0.35, 0.99, 3.99, 3.67, 2.07, 0.96, 5.35, 2.90, 13.77
38	8	0.47, 0.73, 1.40, 0.74, 0.39, 1.13, 0.09, 2.38

- Key result: **Higher voltage $\rightarrow$ shorter breakdown times.**
- If testing had stopped at 180 minutes, some observations would have been **censored** (true failure times unknown but $>$180 minutes).

Example 1.1.7 – Clinical Trial (Leukemia)

- Gehan (1965) compared **6-mercaptopurine (6-MP)** with placebo in acute leukemia patients.
- Outcome: **remission time** – the duration patients remained in remission (a state where symptoms reduce or disappear, but the disease may return later).
- Two groups of 21 patients each (placebo vs. 6-MP).

Table 1.3. Lengths of Remission (in weeks) for Two Groups of Patients[a]

6-MP	6, 6, 6, 6*, 7, 9*, 10, 10*, 11*, 13, 16, 17*, 19*, 20*, 22, 23, 25*, 32*, 32*, 34*, 35*
Placebo	1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23

[a] Stars denote censored observations.

Subsection 2

1.2 Lifetime Distributions

Cumulative Distribution Function (CDF)

- Let T be a **nonnegative random variable** representing lifetimes of individuals in a population.
- The probability density function (*pdf*) of T is denoted by $f(t)$, and the **cumulative distribution function** (*cdf*) of T is defined as

$$F(t) = \Pr(T \leq t) = \int_0^t f(x)\,dx, \quad t \geq 0$$

Survivor Function

- The probability that an individual survives **beyond** time t is

$$S(t) = \Pr(T > t) = \int_t^\infty f(x)\,dx$$

- In reliability engineering (e.g., lifetimes of machines), $S(t)$ is also called the **reliability function**.
- Properties of $S(t)$:
  - $S(0) = 1$ (everyone is alive at the start).
  - $S(t)$ is monotone decreasing.
  - $\lim_{t\to\infty} S(t) = 0$.

Survivor Function

- A useful relationship between mean survival and the survivor function:

$$E(T) = \int_0^\infty S(x)\,dx$$

This shows that the **mean survival time** equals the area under the survivor curve.

Quantiles

- The pth quantile of T is the time t_p such that

$$F(t_p) = \Pr(T \leq t_p) = p \quad \Rightarrow \quad t_p = F^{-1}(p) = S^{-1}(1-p)$$

- Special cases:
  - t_p is also the $100p^{\text{th}}$ percentile.
  - The 0.5 quantile ($t_{0.5}$) is the **median lifetime**.

Mortality Rate

- In life tables, the **mortality rate** at time t is the probability of dying between t and $t+1$ among those alive at time t:

$$q_t = \Pr(t \leq T < t+1 \,|\, T \geq t)$$

- Properties:
  - Mortality rate is a probability, so $0 \leq q_t \leq 1$.
  - As the interval shrinks, mortality rate leads to the **hazard function**.

Hazard Function

- Defined as the **instantaneous failure rate** at time t, conditional on survival up to t:

$$h(t) = \lim_{\Delta t \to 0} \frac{\Pr(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} = \frac{f(t)}{S(t)}$$

- Key points:
  - Hazard is a **rate, not a probability** $\rightarrow$ it can take values from 0 to ∞.
  - It represents the **instantaneous risk of failure** at time t.

Hazard Function

- Approximation:

$$\Pr(t \leq T < t + \Delta t \mid T \geq t) \approx h(t)\Delta t$$

- Alternative interpretation: for small Δt, $h(t)\,\Delta t$ approximates the **conditional failure probability** over $[t, t + \Delta t)$.
- Other names: *force of mortality*, *conditional failure rate*.

Hazard Function

- Suppose we are told that for a certain interval of length Δt, the probability of failure is

$$P = \Pr(t \leq T < t + \Delta t \mid T \geq t) = \tfrac{1}{4}.$$

- To get the corresponding **hazard rate**, divide by Δt:

$$h(t) \approx \frac{P}{\Delta t}$$

Then the hazard depends on the interval length:

P	Δt	$P/\Delta t$ (rate)
$1/4$	$1/3$ day	0.75/day
$1/4$	$1/21$ week	5.25/week

Relationship Between Different Functions

- The four functions $f(t)$, $F(t)$, $S(t)$, and $h(t)$ are mathematically equivalent ways of describing the distribution of T.
- From a given expression of one function, say hazard function, expressions of other functions (e.g. density function) can be derived

Relationship Between Different Functions

Expressing $S(t)$ in terms of $h(t)$

$$h(x) = \frac{f(x)}{S(x)} = -\frac{d}{dx}\log S(x)$$

$$\int_0^t h(x)\,dx = \int_0^t \Big[-\frac{d}{dx}\log S(x)\Big]dx$$

$$-\int_0^t h(x)\,dx = \log S(x)\Big|_0^t$$

$$-\int_0^t h(x)\,dx = \log S(t) - \log S(0)$$

Since $S(0) = 1$,

$$S(t) = \exp\Big(-\int_0^t h(x)\,dx\Big)$$

Cumulative Hazard Function

- It is useful to define the *cumulative hazard function* as

$$H(t) = \int_0^t h(x)\,dx$$

- Relationship with survivor function:

$$S(t) = \exp[-H(t)] \quad \Rightarrow \quad H(t) = -\log S(t)$$

- Notes:
  - $S(\infty) = 0 \Rightarrow H(\infty) = \infty$.
  - It is possible for $H(t) > 1$ (since it is not a probability).

Cumulative Hazard Function

- For a given time t, the greater the risk, the smaller $S(t)$, and hence the shorter mean survival time $E(T)$, and vice verse
- It is possible for the cumulative hazard function to exceed unity

$$H(t) \geq 1 \;\Rightarrow\; -\log S(t) \geq 1 \;\Rightarrow\; S(t) \leq e^{-1} = 0.368$$

- The cumulative hazard is then greater than unity when the probability of an event occurring after time t is less than 0.37

Relationship Between Different Functions

Expressing $f(t)$ in terms of $h(t)$

$$h(t) = \frac{f(t)}{S(t)}$$

$$f(t) = h(t)S(t) = h(t)\,\exp\Big(-\int_0^t h(x)\,dx\Big)$$

Example 1.2.1

- Suppose T has pdf

$$f(t) = \beta t^{\beta-1} \exp(-t^{\beta}), \quad t > 0$$

  - Obtain survivor function and hazard function of T

Discrete Models: Motivation

- Sometimes lifetimes are measured in **cycles or intervals** (e.g., weeks, months, visits).
- Then T is a **discrete random variable**.
- Possible values: $t_1, t_2, ...,$ with $0 = t_0 < t_1 < t_2 < \cdots$.

Discrete Probability Functions

- **Probability mass function (pmf):**

$$f(t_j) = \Pr(T = t_j), \quad j = 1, 2, ...$$

- **Survivor function:**

$$S(t) = \Pr(T \geq t) = \sum_{j: t_j \geq t} f(t_j)$$

- Properties:
  - Step function (non-increasing).
  - Left-continuous.
  - $S(0) = 1,\ S(\infty) = 0.$

Note: In continuous-time settings we often write $S(t) = \Pr(T > t)$ (right-continuous); here, with discrete times, $S(t) = \Pr(T \geq t)$ is left-continuous.

Discrete Hazard Function

- Defined as
$$h(t_j) = \Pr(T = t_j \mid T \geq t_j) = \frac{f(t_j)}{S(t_j)}.$$
- Interpretation: probability of failing **at time** t_j given survival up to that time.
- Range: $0 \leq h(t_j) \leq 1$.
- Unlike continuous hazard, this is a **probability, not rate**.

Discrete Hazard Function

- Connection to survivor:

$$h(t_j) = 1 - \frac{S(t_{j+1})}{S(t_j)}.$$

- since $f(t_j) = S(t_j) - S(t_{j+1})$

- Survivor in terms of hazards:

$$S(t) = \prod_{t_j < t} [1 - h(t_j)].$$

Intuition

To survive until t_j: must "not fail" at earlier times $[1 - h(t_1)], [1 - h(t_2)], \ldots, [1 - h(t_{j-1})]$.

Discrete Hazard Function

- pmf in terms of hazards:

$$f(t_j) = h(t_j) \prod_{i=1}^{j-1} [1 - h(t_i)].$$

Intuition

To fail exactly at t_j: multiply survival so far by hazard at t_j.

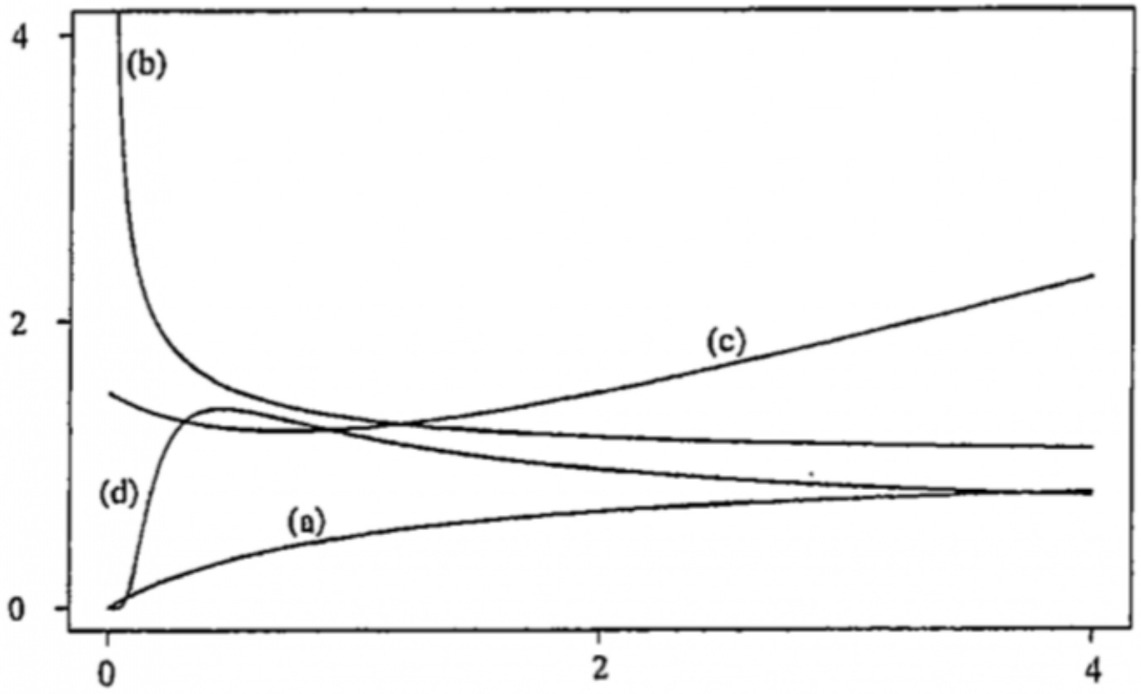
Some Remarks on Hazard Functions

- The hazard function is an important characteristic of a lifetime distribution that indicates the way the risk of failure varies with age or time, and this is of interest in most applications.
- In many instances, information is available on how failure rates change with time and such prior information about the shape of the hazard function can help guide model selection.
- The model/information for hazard function can easily be translated for survivor and density functions using the formulas derived earlier

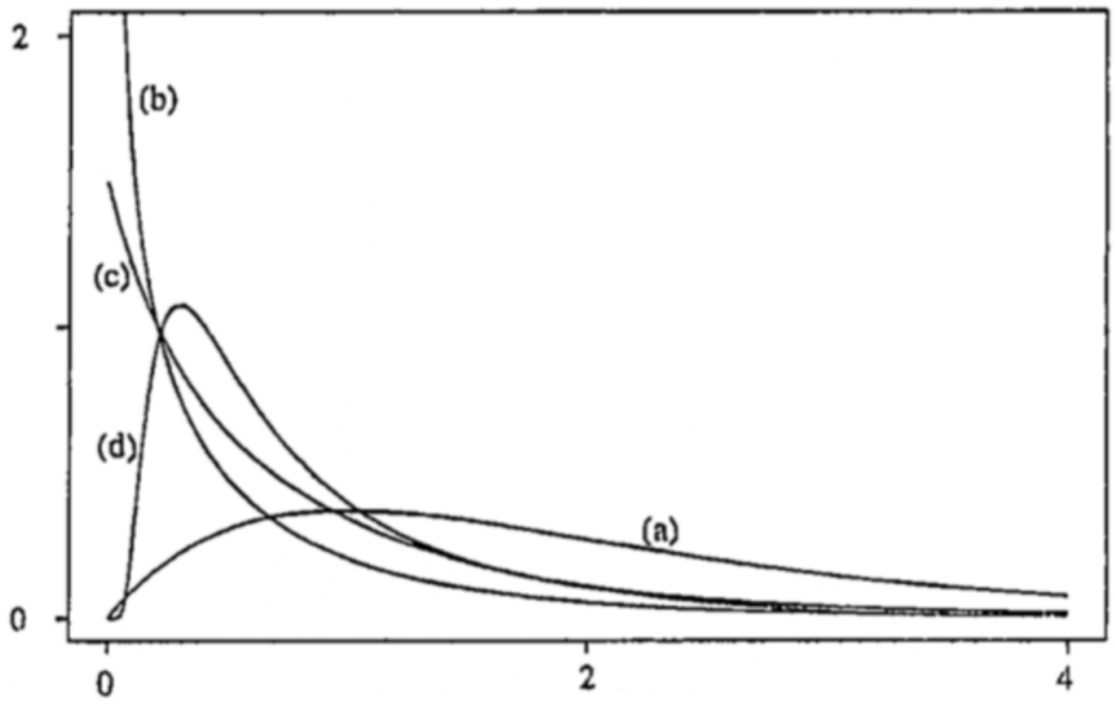
Different Shapes of Hazard Functions

- The shapes of hazard functions could be different, such as
  - monotone increasing (e.g. positive aging) (a)
  - monotone decreasing (e.g. negative aging) (b)
  - bathtub-shaped or U-shaped (e.g. age at death of human populations, lifetime of manufactured items, etc.) (c)
  - inverse bathtub-shaped (e.g. survival after treatment for cancer, duration of marriage, etc.) (d)

Different Shapes of Hazard Functions



Different Shapes of Density Functions



Some Remarks on Hazard Functions

- Shapes of density function could be different corresponding to the shapes of hazard functions
- Although different survivor functions can have the same basic shape, their hazard functions can differ dramatically
- The hazard function is usually more informative about the underlying mechanism of failure than the survivor function.
- Modelling the hazard function is an important method for summarizing survival data

Subsection 3

1.3 Some Important Failure Time Models

Introduction

- In survival analysis, **parametric models** are often used to describe lifetime data.
- Only a few distributions have proven useful across many applications.
- The most common univariate models are:
  - Exponential
  - Weibull
  - Log-normal
  - Log-logistic
- Notations
  - $T \rightarrow$ lifetime, takes only nonnegative values, i.e. from 0 to ∞
  - $Y = \log T \rightarrow$ log-lifetime, takes any value on the real line, i.e. from $-\infty$ to ∞

The Exponential Distribution

- The exponential distribution is characterized by a constant **hazard function**

$$h(t) = \lambda, \; t \geq 0$$

  - $\lambda > 0$

- The **cumulative hazard function**

$$H(t) = \int_0^t h(x)\, dx = \int_0^t \lambda\, dx = \lambda t$$

- The **survivor function**

$$\begin{aligned} S(t) &= \exp(-H(t)) \\ &= \exp(-\lambda t) \end{aligned}$$

The Exponential Distribution

- The **probability density function**

$$f(t) = h(t)\, S(t) = \lambda \exp(-\lambda t)$$

- Reparameterization $\theta = \lambda^{-1}$ Then, $T \sim \text{Exp}(\text{scale} = \theta)$, where

$$f(t) = (1/\theta)\, \exp(-t/\theta), \;\; t \geq 0$$

The Exponential Distribution

- Properties
  - $E(T) = \theta$
  - $V(T) = \theta^2$
- **Quantiles**, the pth quantile

$$F(t_p) = p \;\Rightarrow\; 1 - \exp(-t_p/\theta) = p$$
$$\Rightarrow t_p = -\theta \log(1-p)$$

  - The median, $.5th$ quantile

$$t_{.5} = -\theta \log(.5)$$

The Exponential Distribution

- The exponential distribution with $\theta = 1$ is known as standard exponential distribution
- If $T \sim \text{Exp}(\theta)$ then
$$(T/\theta) \sim \text{Exp}(1)$$
  - The mean and variance of $\text{Exp}(1)$ is 1
  - The median of the $\text{Exp}(1)$ is $-\log(.5) = 0.6931$
  - The density function of $\text{Exp}(1)$ is positively skewed

The Exponential Distribution

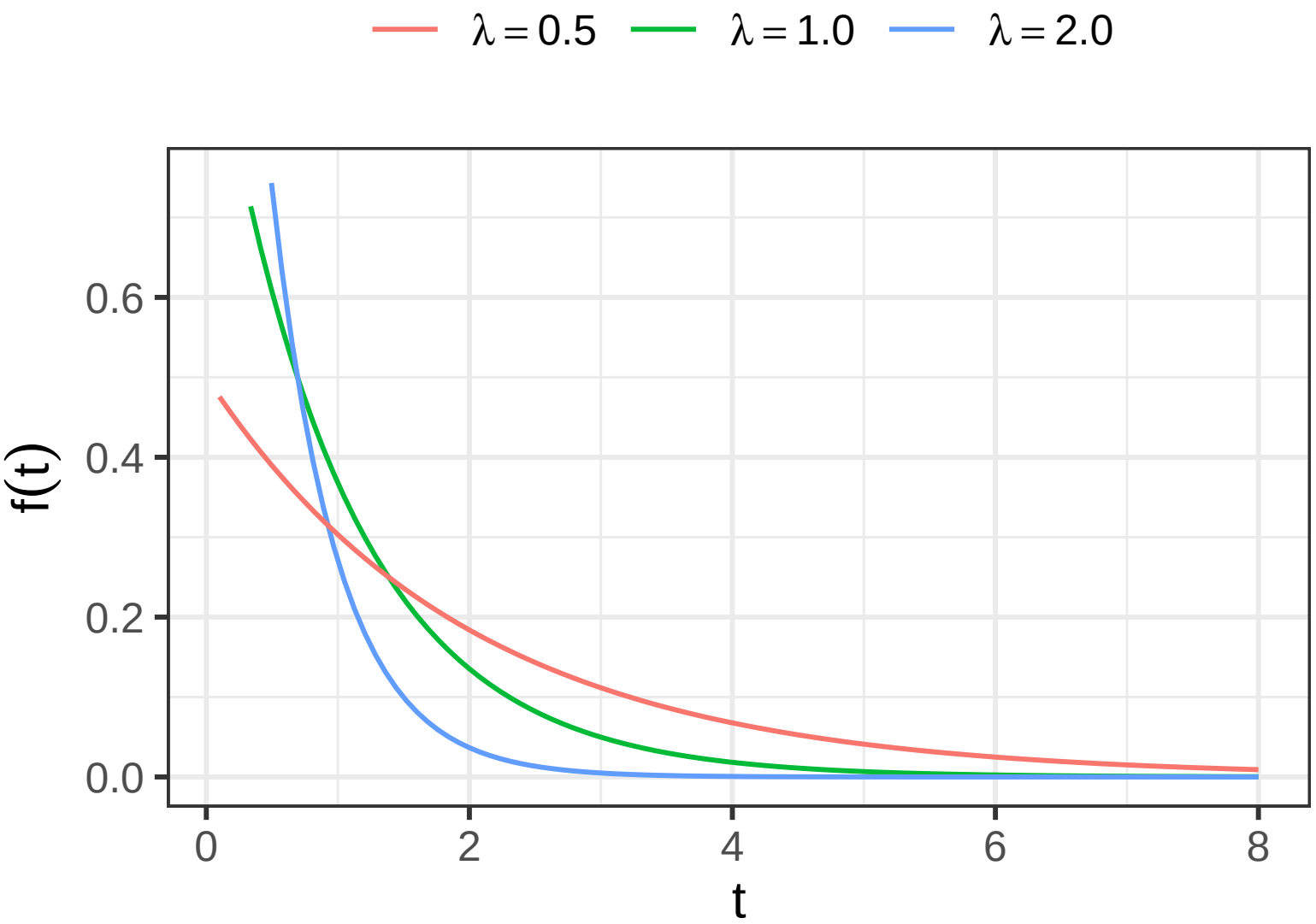


Figure 2: Density function of exponential distribution

The Exponential Distribution

- Historically, the exponential was the first widely discussed lifetime distribution model
  - This was in part because of the availability of simple statistical methods for it
- The assumption of a constant hazard function is very restrictive, so the model's applicability is fairly limited

The Weibull Distribution

- The Weibull distribution is the most widely used lifetime distribution model.
- It has applications to the lifetimes or durability of manufactured
  - It is used as a model with diverse types of items, such as ball bearings, automobile components, and electrical insulation.
- It is also used in biological and medical applications, for example, in studies on the time to the occurrence of tumors in human populations or in laboratory animals.

The Weibull Distribution

- The hazard function of Weibull distribution

$$h(t) = \lambda\beta(\lambda t)^{\beta-1},\ \lambda > 0,\ \beta > 0.$$

- Show that $h(t)$ is
  1. monotone increasing for $\beta > 1$
  2. monotone decreasing for $\beta < 1$
  3. constant for $\beta = 1$
- Exponential distribution is a special case
  - For $\beta = 1$, Weibull distribution reduces to exponential distribution with $h(t) = \lambda$

The Weibull Distribution

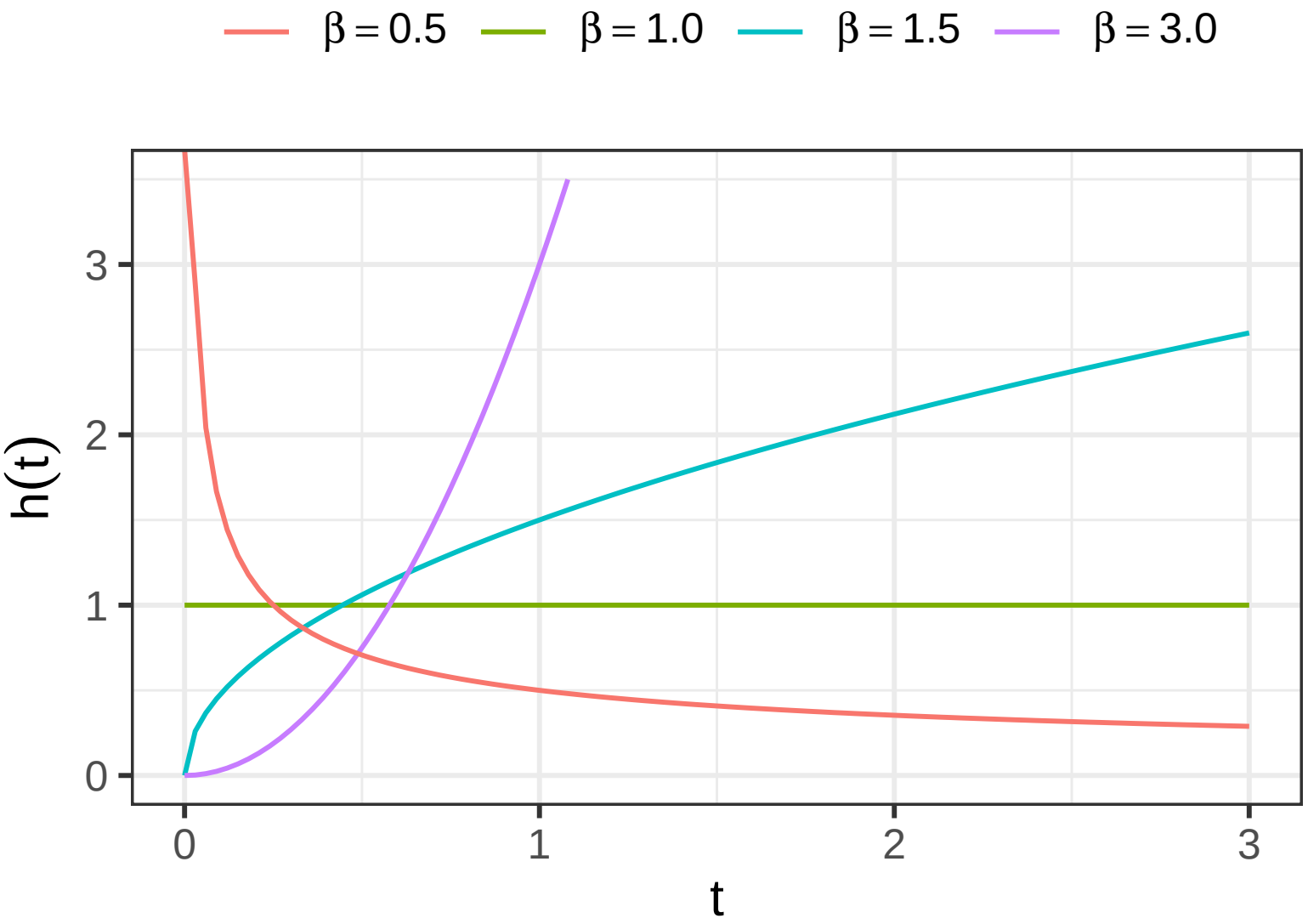


Figure 3: Hazard function of Weibull distribution $(\lambda = 1.0)$

The Weibull Distribution

- The cumulative hazard function

$$H(t) = \int_0^t h(x)\,dx = \int_0^t \lambda\beta\,(\lambda x)^{\beta-1}\,dx = (\lambda\,t)^{\beta}$$

- The survivor function

$$S(t) = \exp[-H(t)] = \exp[-(\lambda t)^{\beta}]$$

- The density function

$$f(t) = h(t)\,S(t) = \lambda\beta\,(\lambda t)^{\beta-1}\exp[-(\lambda t)^{\beta}]$$

The Weibull Distribution

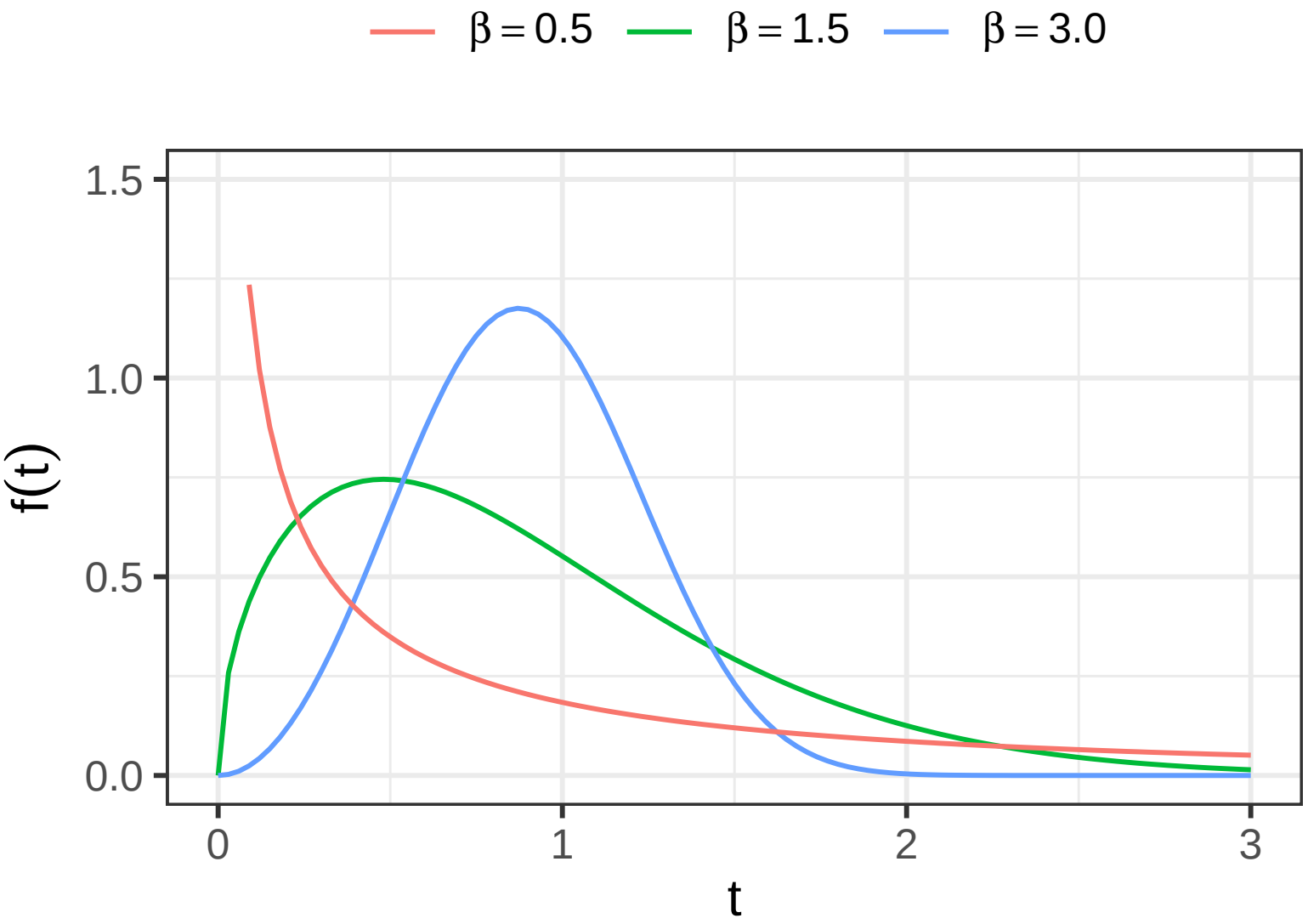


Figure 4: Density function of Weibull distribution $(\lambda = 1.0)$

The Weibull Distribution

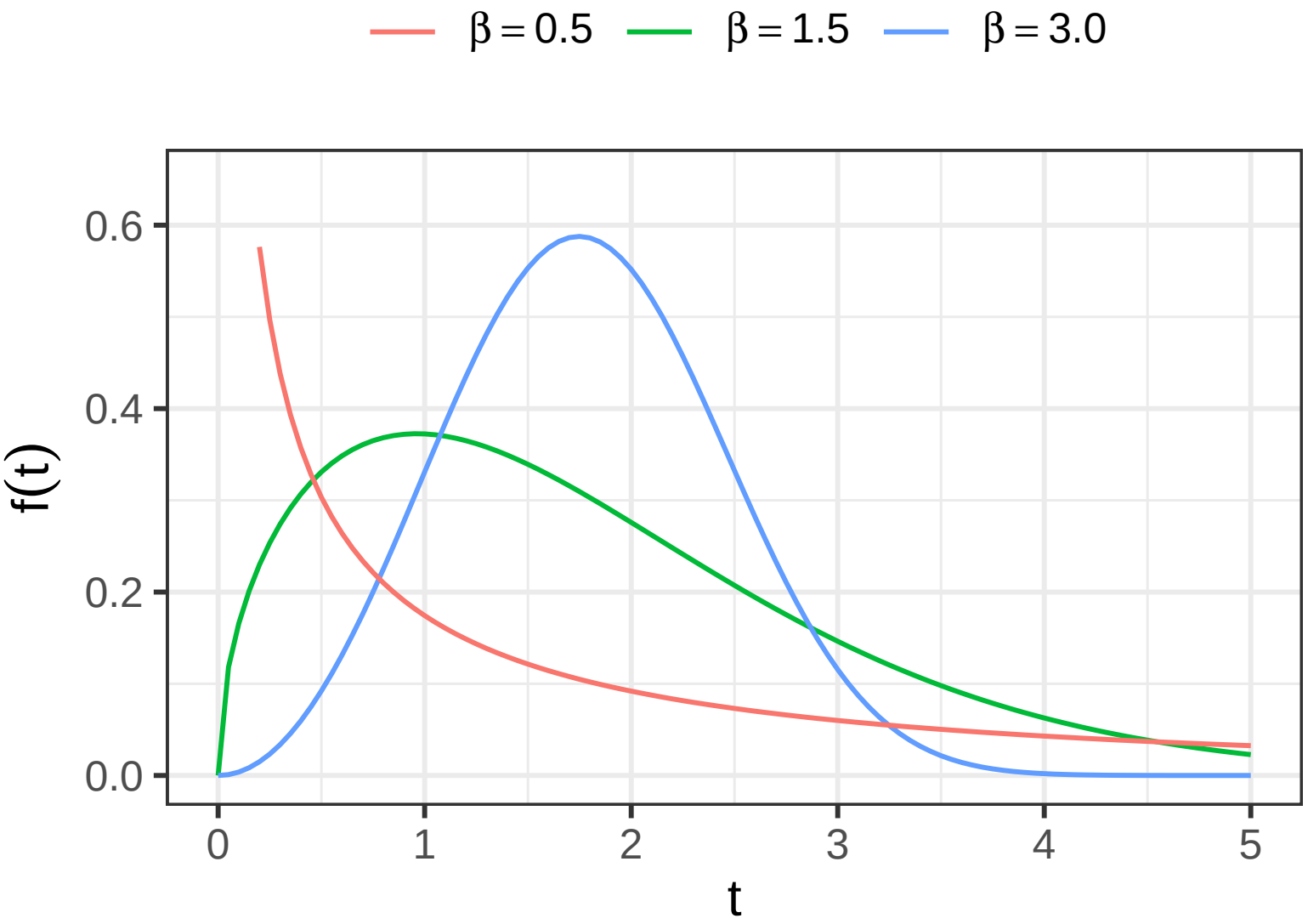


Figure 5: Density function of Weibull distribution $(\lambda = 2.0)$

The Weibull Distribution

- Show that the rth moment of Weibull distribution

$$E(T^r) = \lambda^{-r}\Gamma(1 + r/\beta)$$

  - Obtain the expressions of $E(T)$ and $V(T)$

The Weibull Distribution

- The pth quantile can be obtained as

$$F(t) = 1 - e^{-(\lambda t_p)^\beta} = p$$
$$\Rightarrow \; t_p = \alpha[-\log(1-p)]^{1/\beta}, \qquad \text{where } \alpha = 1/\lambda$$

  - $\alpha = 1/\lambda$ is known as the scale parameter of the distribution
  - The shape of the distribution depends on β, which is known as the shape parameter
- It can be shown that α is the .632 quantile of the distribution irrespective of the value of β
  - i.e. α is greater than the median of the distribution!

The Extreme Value Distribution

- Let T follows a Weibull distribution

$$T \sim \text{Weib}(\alpha, \beta) \;\; \text{with} \;\; \alpha = 1/\lambda$$

- Extreme value distribution (also known as Gumbel distribution) is closely related to Weibull distribution
- If lifetime T follows a Weibull distribution then log-lifetime $Y = \log T$ follows an extreme value distribution
- Extreme value distribution has two parameters, which have one-to-one connection with the Weibull distribution parameters!

The Extreme Value Distribution

- **Exercise**: If $T \sim$ Weib(α, β), obtain the pdf of $Y = \log T$
  - *Hints.* $J = \frac{dt}{dy} = e^y$ and

$$f_Y(y) = f_T(e^y) \mid J \mid$$

The Extreme Value Distribution

- $T \sim \text{Weib}(\alpha, \beta) \quad \Leftrightarrow \quad Y = \log T \sim \text{EV}(u, b)$
  - $u = \log \alpha$ and
  - $b = (1/\beta)$
- The pdf of Y

$$f(y) = (1/b) \exp\left[\frac{y-u}{b} - \exp\left(\frac{y-u}{b}\right)\right] \quad -\infty < y < \infty$$

  - $-\infty < u < \infty$ and $b > 0$

The Extreme Value Distribution

- The survivor function

$$
\begin{aligned}
S(y) &= \int_y^{\infty} f(x)\,dx \\
&= \int_y^{\infty} (1/b)\exp\left[\frac{x-u}{b} - \exp\left(\frac{x-u}{b}\right)\right] dx \\
&= \exp\left[-\exp\left(\frac{y-u}{b}\right)\right]
\end{aligned}
$$

- The cumulative hazard function

$$
H(y) = \exp\left(\frac{y-u}{b}\right)
$$

- The hazard function

$$
h(y) = \frac{d\,H(y)}{dy} = (1/b)\exp\left(\frac{y-u}{b}\right)
$$

The Extreme Value Distribution

Standard extreme value distribution

- If $Y \sim \mathsf{EV}(u, b)$, then

$$\frac{Y - u}{b} \sim \mathsf{EV}(0, 1),$$

the *standard extreme value distribution*.

The Extreme Value Distribution

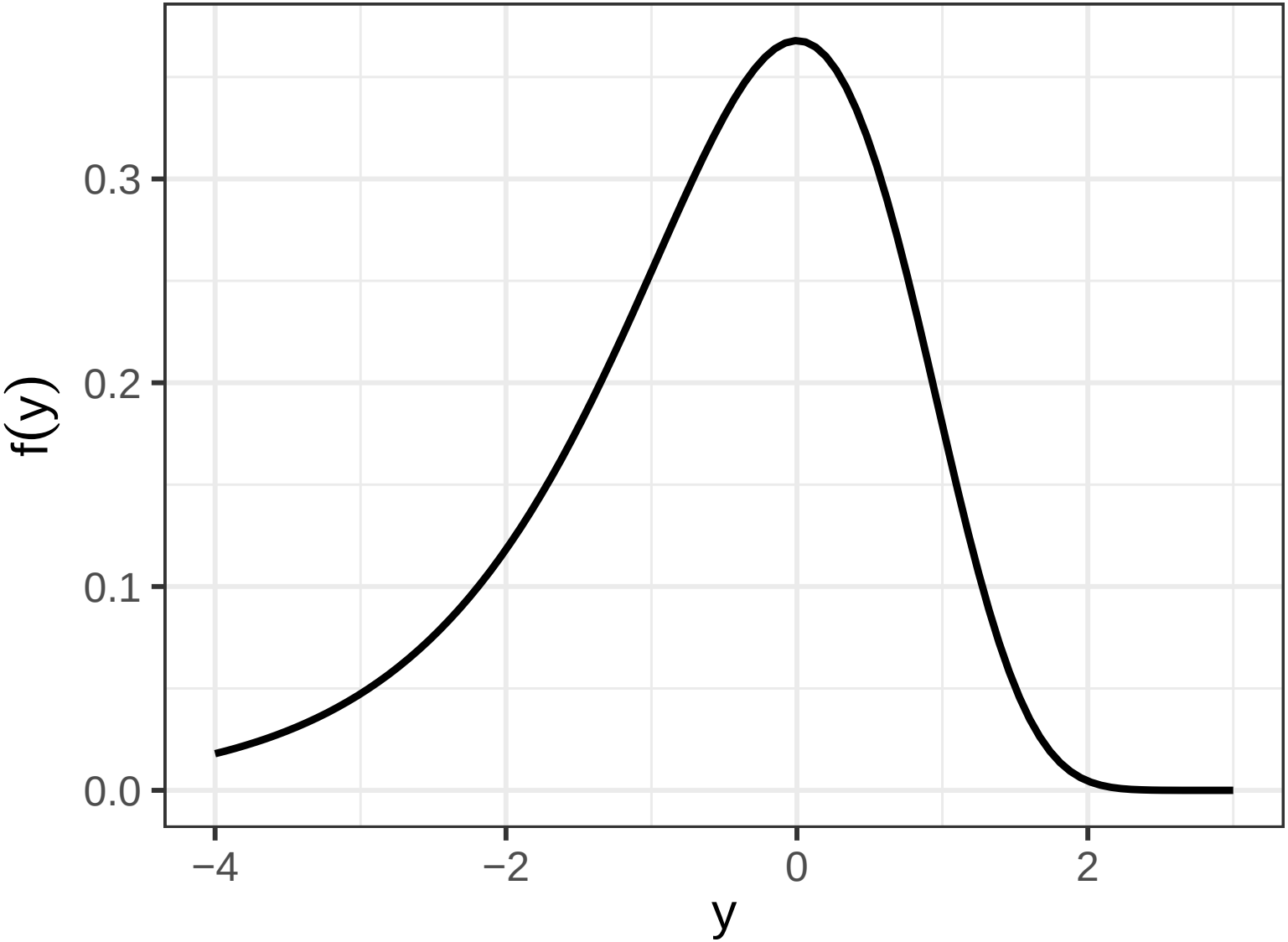


Figure 6: The density function of extreme value distribution with $u = 0$ and $b = 1$

The Extreme Value Distribution

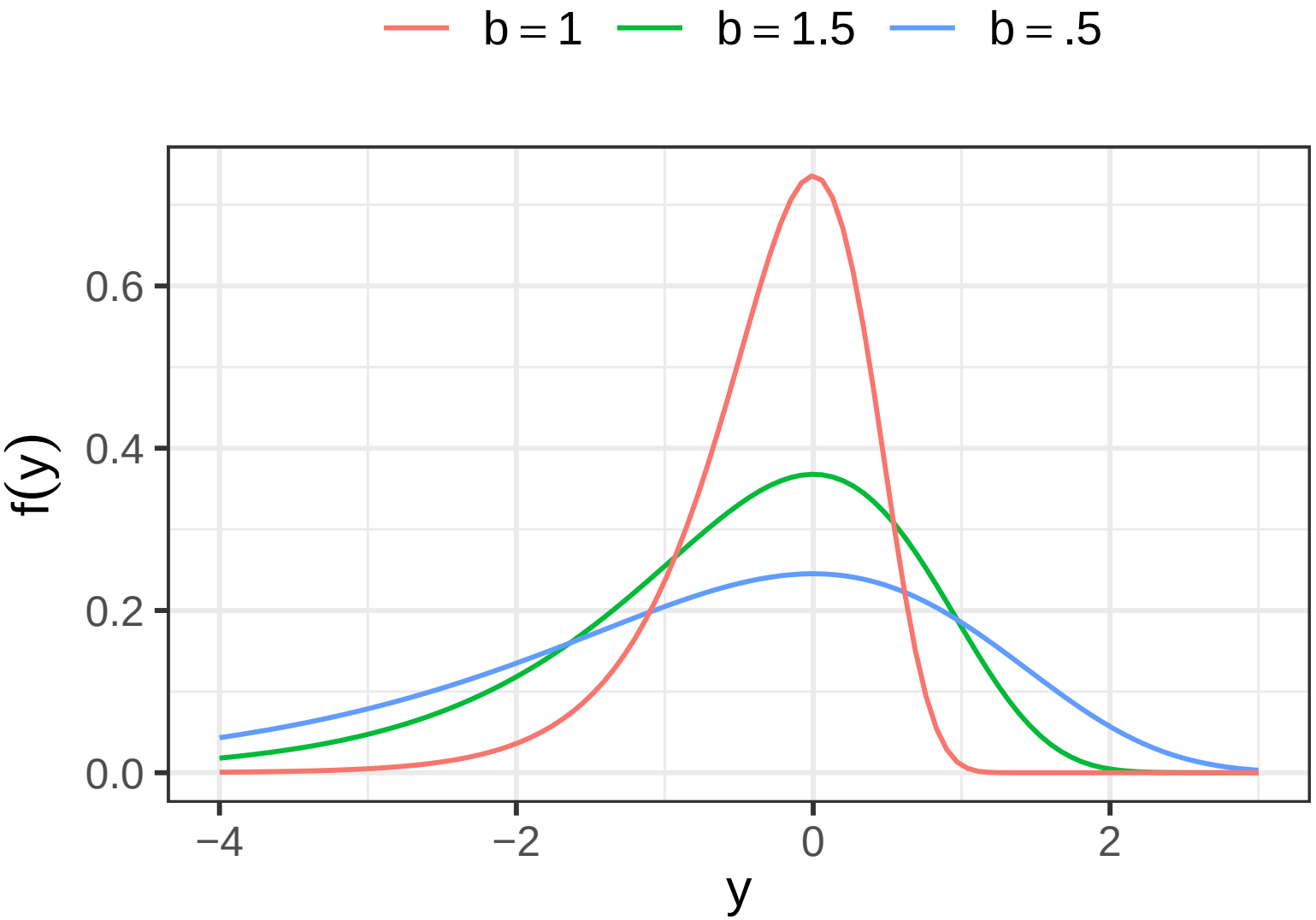


Figure 7: The density function of extreme value distribution with $u = 0$

The Extreme Value Distribution

- The moment generating function of $EV(u, b)$

$$M_Y(\theta) = E[e^{\theta Y}] = \int_{-\infty}^{\infty} e^{\theta y} f(y)\, dy$$

$$= \int_{-\infty}^{\infty} e^{\theta y} (1/b) \exp\left[\frac{y-u}{b} - \exp\left(\frac{y-u}{b}\right)\right] dy$$

$$= \int_{-\infty}^{\infty} e^{\theta(u+bz)} \exp\left[z - \exp(z)\right] dz \qquad [\text{let } \frac{y-u}{b} = z]$$

$$= \int_{0}^{\infty} e^{\theta u}\, x^{\theta b}\, e^{-x}\, dx = e^{\theta u}\, \Gamma(\theta b + 1), \qquad [\text{let } e^z = x]$$

The Extreme Value Distribution

- If $Y \sim EV(u, b)$

$$M(\theta) = e^{\theta u}\,\Gamma(\theta b + 1)$$

- If $Y \sim EV(0, 1)$

$$M(\theta) = \Gamma(\theta + 1)$$

The Extreme Value Distribution

- Moments of standard extreme value distribution $Z \sim EV(0,1)$

$$E(Z) = \frac{d}{d\theta} M(\theta)\Big|_{\theta=0} = \Gamma'(1) = -\gamma \ \ \text{(Euler's constant)}$$
$$V(Z) = \Gamma''(1) - \gamma^2 = \pi^2/6$$

- For $Y \sim EV(u,b)$, show that

$$E(Y) = u - \gamma b \ \ \text{and} \ \ V(Y) = b^2(\pi^2/6)$$

The Extreme Value Distribution

The pth quantile of extreme value distribution

$$
\begin{aligned}
F(y_p) &= p \\
S(y_p) &= 1 - p \\
\exp\left[-\exp\left(\frac{y_p - u}{b}\right)\right] &= (1-p) \\
y_p &= u + b\log\left[-\log(1-p)\right]
\end{aligned}
$$

- Show that the location parameter u is the .632 quantile of $Y \sim EV(u, b)$

The Log-normal Distribution

- The lifetime T is said to be log-normally distributed if log-lifetime $Y = \log T$ is normally distributed.
- The parameters of normal distribution μ and σ are also considered as the parameters of log-normal distribution

$$\begin{aligned} Y &= \log T \sim N(\mu, \sigma^2) \\ \Rightarrow\ T &= \exp(Y) \sim \log N(\mu, \sigma^2) \end{aligned}$$

The Log-normal Distribution

- Let $Y = \log T \sim N(\mu, \sigma^2)$, show that the density function of $T = \exp(Y)$ is

$$f_T(t) = f_Y(\log T)\left|\frac{dy}{dt}\right|$$
$$= \frac{1}{\sigma t\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\log t - \mu}{\sigma}\right)^2\right]$$

  - $t > 0$, $\sigma > 0$, and $-\infty < \mu < \infty$

The Log-normal Distribution

- The survivor function of $T = \exp(Y)$

$$S(t) = 1 - \Phi\left(\frac{\log t - \mu}{\sigma}\right)$$

  - $\Phi(\cdot)$ $\rightarrow$ distribution function of $N(0, 1)$

- The hazard function is defined as $f(t)/S(t)$, which takes the value 0 at $t = 0$, increases to a maximum and then decreases, approaching 0 as $t \rightarrow \infty$.

The Log-normal Distribution

- It can be shown

$$E(T) = \exp(\mu + \sigma^2/2)$$
$$V(T) = [\exp(\sigma^2) - 1][\exp(2\mu + \sigma^2)]$$

- For log-normal distribution
  - $\exp(\mu)$ $\rightarrow$ the scale parameter
  - $1/\sigma$ $\rightarrow$ the shape parameter
- Show that for $T \sim \log N(\mu, \sigma^2)$

$$t_{.5} = \exp(\mu)$$

The Log-normal Distribution

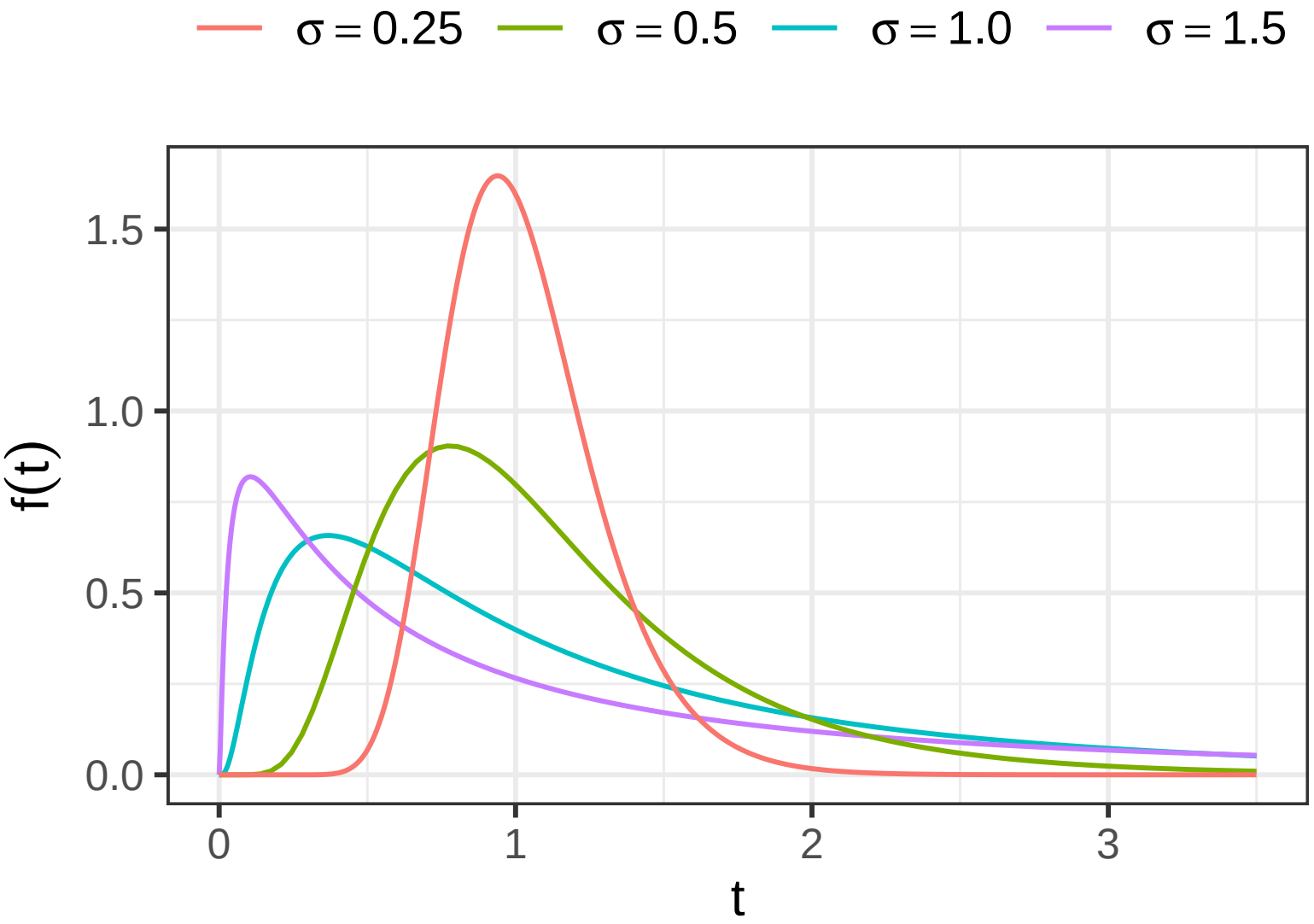


Figure 8: Density function of log-normal distribution with $\mu = 0$

The Log-normal Distribution

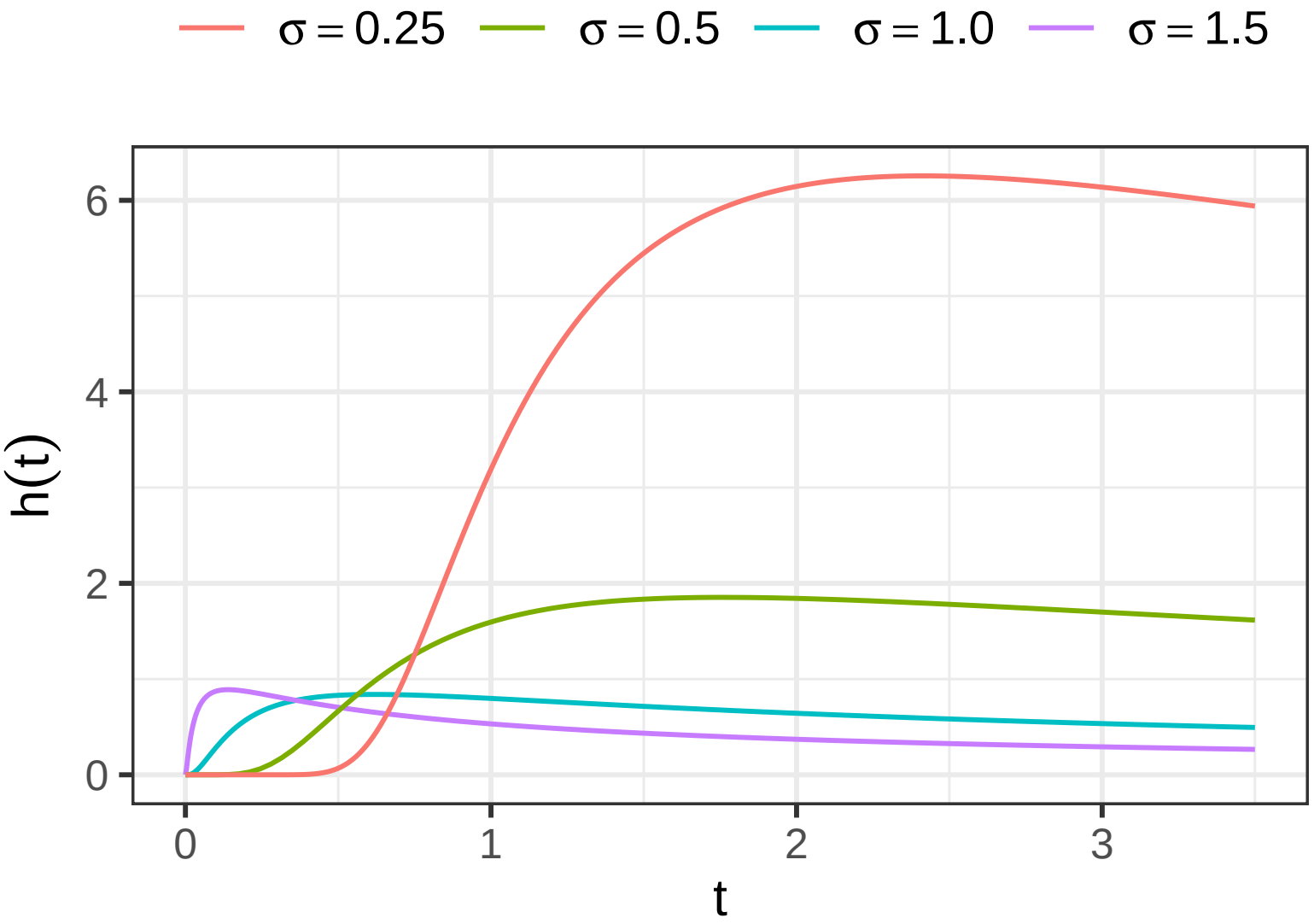


Figure 9: Hazard function of log-normal distribution with $\mu = 0$

The Log-logistic distribution

- If $Y = \log T$ follows a logistic distribution then T follows a log-logistic distribution
- The probability density function of a logistic distribution is a unimodal, bell-shaped curve symmetric around its mean, and for location parameter u and scale parameter b it is given by

$$f(y) = \frac{e^{(y-u)/b}}{b\left[1 + e^{(y-u)/b}\right]^2}.$$

- $-\infty < y < \infty$, $-\infty < u < \infty$, $b > 0$

The Log-logistic distribution

- The survivor function of a logistic distribution is

$$S(y) = \int_y^\infty f(x)\,dx = \int_y^\infty \frac{e^{(x-u)/b}}{b\left[1+e^{(x-u)/b}\right]^2}\,dx.$$
$$= \cdots$$
$$= \frac{1}{\left[1+e^{(y-u)/b}\right]}$$

- The hazard function of logistic distribution is

$$h(y) = \frac{f(y)}{S(y)} = \frac{e^{(y-u)/b}}{b\left[1+e^{(y-u)/b}\right]}.$$

The Log-logistic distribution

- The pdf of log-logistic distribution

$$
\begin{aligned}
f_T(t) &= f_Y(\log T)\left|\frac{dy}{dt}\right| \\
&= \frac{\beta}{\alpha}\frac{(t/\alpha)^{\beta-1}}{\left[1+(t/\alpha)^{\beta}\right]^2}
\end{aligned}
$$

  - $\alpha = \exp(u)$ and $\beta = 1/b$

The Log-logistic distribution

- The survivor function of $T \sim \text{LLogis}(\alpha, \beta)$ is

$$S(t) = \int_t^\infty \frac{\beta}{\alpha} \frac{(x/\alpha)^{\beta-1}}{\left[1 + (x/\alpha)^\beta\right]^2} \, dx$$

  - Let $(x/\alpha)^\beta = y$

$$\begin{aligned} S(t) &= \int_{(t/\alpha)^\beta}^\infty \frac{1}{(1+y)^2} \, dy \\ &= \left. \frac{-1}{1+y} \right|_{(t/\alpha)^\beta}^\infty \\ &= \left[1 + (t/\alpha)^\beta\right]^{-1} \end{aligned}$$

The Log-logistic distribution

- The pdf

$$f(t) = \frac{\beta}{\alpha} \frac{(t/\alpha)^{\beta-1}}{\left[1 + (t/\alpha)^{\beta}\right]^2}$$

- The survivor function

$$S(t) = \left[1 + (t/\alpha)^{\beta}\right]^{-1}$$

- The hazard function

$$h(t) = \frac{\beta}{\alpha} \frac{(t/\alpha)^{\beta-1}}{\left[1 + (t/\alpha)^{\beta}\right]}$$

The Log-logistic distribution

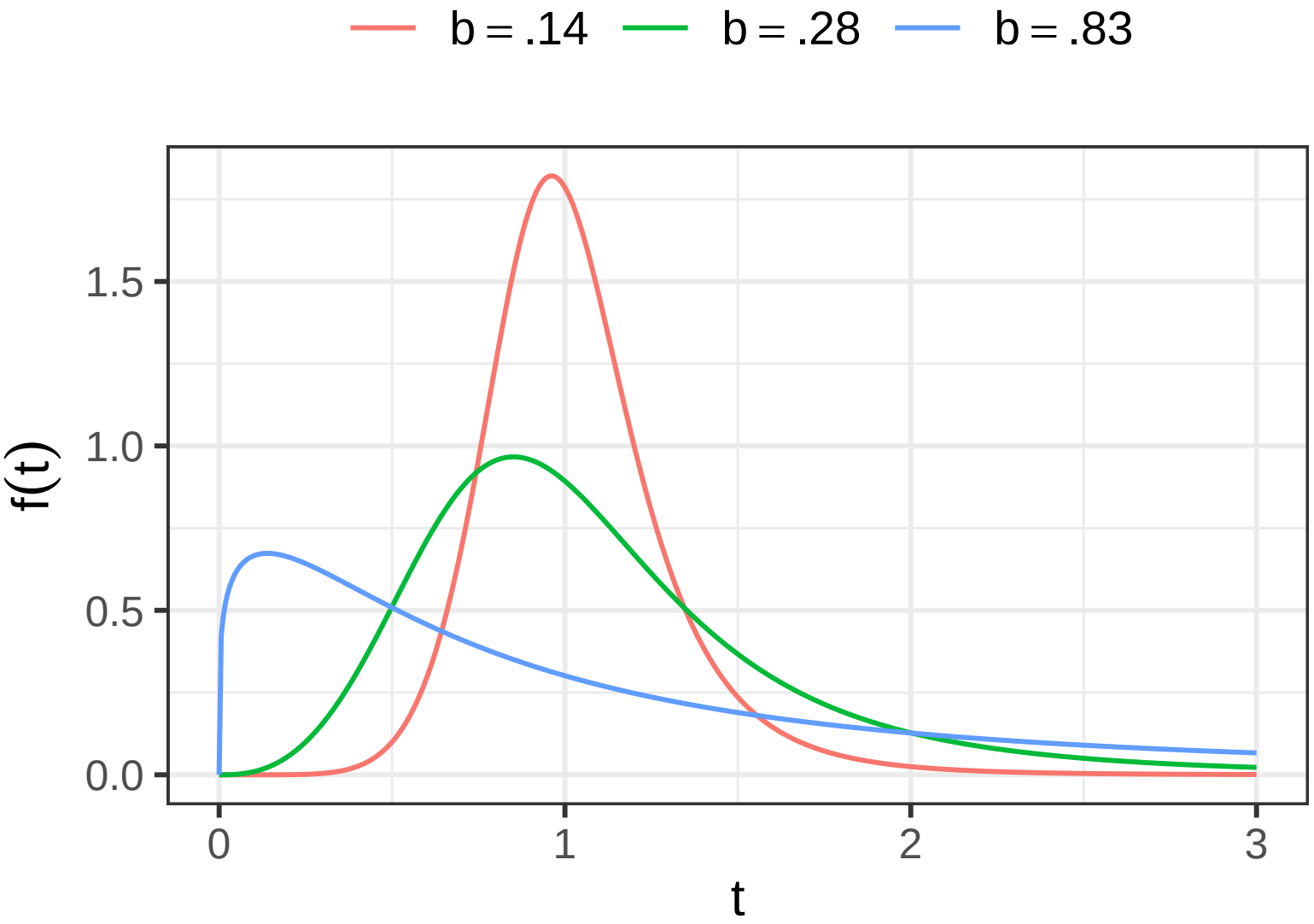


Figure 10: The density function of log-logistic distribution with $u = 0$

The Log-logistic distribution

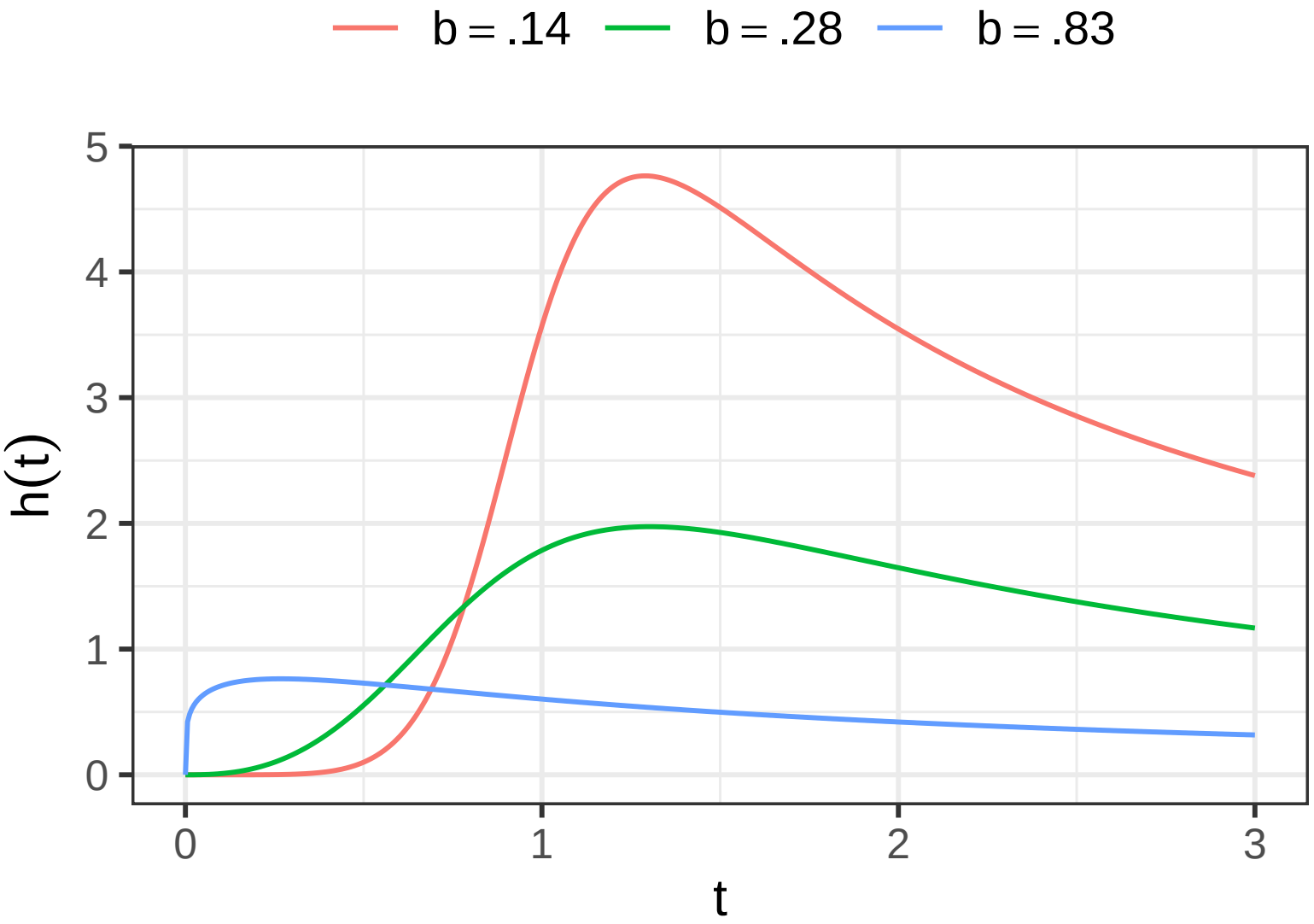


Figure 11: The hazard function of log-logistic distribution with $u = 0$

The Log-logistic distribution

- Show that for $T \sim \text{LLogis}(\alpha, \beta)$, provided $\beta > r$

$$E(T^r) = \int_0^\infty t^r \, \frac{\beta}{\alpha} \, \frac{(t/\alpha)^{\beta-1}}{\left(1 + (t/\alpha)^\beta\right)^2} \, dt.$$

$$= \cdots$$

$$= \alpha^r \, \Gamma\left(\frac{r}{\beta} + 1\right) \Gamma\left(1 - \frac{r}{\beta}\right)$$

You might have to recognize and use a Beta integral.

- Beta (first kind): for $0 < x < 1$,

$$\int_0^1 x^{\alpha-1}(1-x)^{\beta-1} \, dx = \mathrm{B}(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

- Beta function (second kind / Beta prime):

$$\int_0^\infty \frac{x^{a-1}}{(1+x)^{a+b}} \, dx = \frac{\Gamma(a)\,\Gamma(b)}{\Gamma(a+b)}, \quad a > 0,\, b > 0.$$

The Log-logistic distribution

Log-logistic hazard shape

- For $\beta > 1$: $h(0) = 0$, increases to a maximum, then decreases to 0 as $t \to \infty$ (unimodal hazard).
- For $\beta = 1$: $h(t) = 1/(\alpha + t)$, strictly decreasing.
- For $0 < \beta < 1$: $h(t)$ is monotone decreasing (starts at ∞).

The Gamma Distribution

- The gamma distribution has a *pdf* of the form

$$f(t) = \frac{\lambda(\lambda t)^{k-1}\, e^{-\lambda t}}{\Gamma(k)} \quad t > 0$$

  - shape $k > 0$
  - rate $\lambda > 0$ (scale $1/\lambda$)
- For $k = 1$, gamma distribution reduces to exponential distribution

The Gamma Distribution

- *Regularized lower incomplete gamma* function

$$I(k, x) = \frac{1}{\Gamma(k)} \int_0^x u^{k-1} \, e^{-u} \, du$$

The Gamma Distribution

- Survivor function

$$S(t) = \int_t^\infty \frac{\lambda(\lambda x)^{k-1}\, e^{-\lambda x}}{\Gamma(k)}\, dx$$

- Let $y = \lambda x$

$$S(t) = \frac{1}{\Gamma(k)} \int_{\lambda t}^\infty y^{k-1}\, e^{-y}\, dy = 1 - I(k, \lambda t)$$

The Gamma Distribution

Gamma hazard shape

The hazard function

$$h(t) = \frac{f(t)}{S(t)}$$

- For $k > 1$: $h(0) = 0$, $h(t)$ increases, $\lim_{t\to\infty} h(t) = \lambda$.
- For $k = 1$: $h(t) \equiv \lambda$.
- For $0 < k < 1$: $h(t)$ decreases, $\lim_{t\to 0} h(t) = \infty$, $\lim_{t\to\infty} h(t) = \lambda$.

In short: Gamma has increasing hazard rate for $k > 1$, constant for $k = 1$, and decreasing hazard rate for $0 < k < 1$

The Gamma Distribution

- The distribution with $\lambda = 1$ is called one-parameter gamma distribution, denoted by $Ga(k)$, and has pdf

$$f(t) = \frac{t^{k-1}\, e^{-t}}{\Gamma(k)} \quad t > 0$$

- If T follows a gamma distribution with scale parameter λ^{-1} and shape parameter k, then show that $\lambda T \sim Ga(k)$
  - *Hints.* $Y = \lambda T$ and $f_Y(y) = f_T(y/\lambda)\,|dt/dy|$

The Gamma Distribution

- If $Y \sim Ga(k)$ then $2Y \sim \chi^2_{(2k)}$
- Let $T_1, \ldots, T_n$ are iid and exponentially distributed with parameter λ
  - $\sum_{i=1}^{n} T_i$ follows a gamma distribution with parameters λ and n
- The moment generating function of $Y \sim$ Ga(k) (rate 1) is

$$M_Y(\theta) = (1-\theta)^{-k}, \quad \theta < 1.$$

The Gamma Distribution

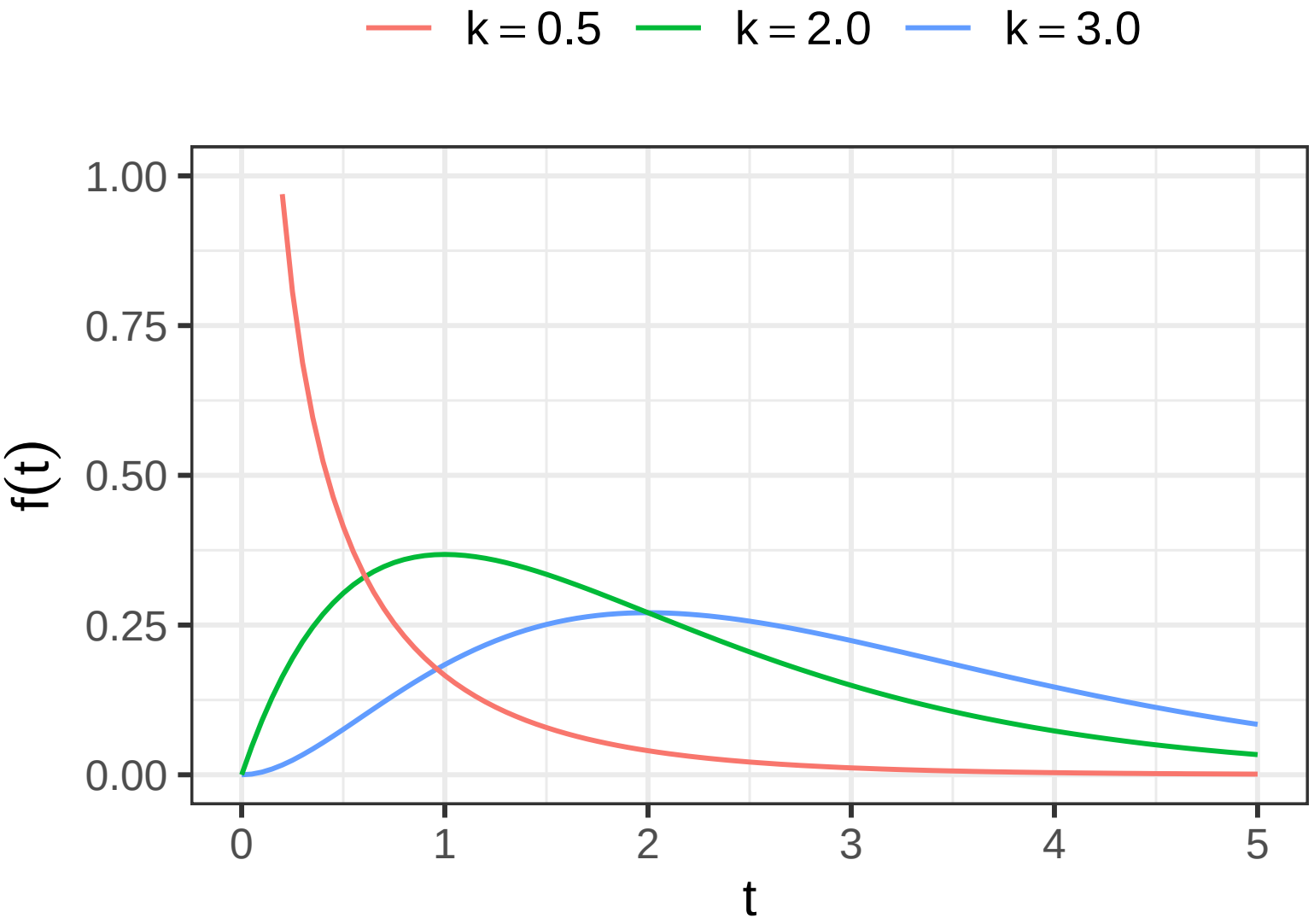


Figure 12: Density function of standard gamma distribution

The Gamma Distribution

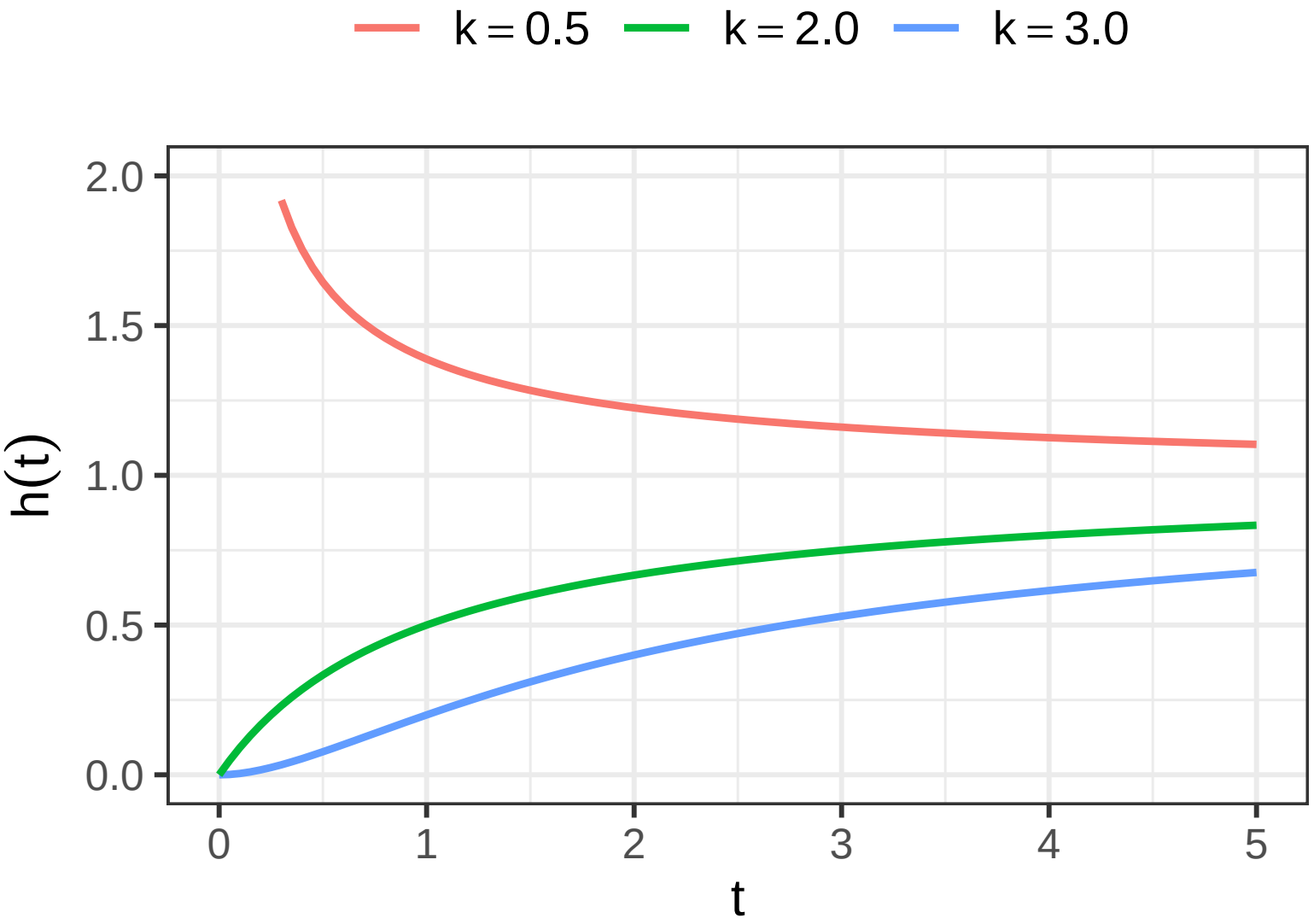


Figure 13: Hazard function of standard gamma distribution

Log-Location-scale Models

Definition: Location Scale Family A location-scale family is a family of distributions formed by *translation* and *rescaling* of a standard family member.

- A parametric location-scale model for a random variable Y is a distribution with pdf of the form

$$f(y) = \frac{1}{b}\, f_0\left(\frac{y-u}{b}\right) \quad -\infty < y < \infty$$

  - $-\infty < u < \infty$, location parameter
  - $b > 0$, scale parameter
  - $f_0(z)$ is a specified pdf on $(-\infty, \infty)$

Log-Location-scale Models

- The cumulative distribution function of Y

$$
\begin{aligned}
F(y) &= \int_{-\infty}^{y} (1/b)\, f_0\left(\frac{x-u}{b}\right) dx \\
&= \int_{-\infty}^{(y-u)/b} f_0(z)\, dz \\
&= F_0\left(\frac{y-u}{b}\right)
\end{aligned}
$$

- Similarly, the survivor function of Y

$$
S(y) = 1 - F_0\left(\frac{y-u}{b}\right) = S_0\left(\frac{y-u}{b}\right)
$$

Log-Location-scale Models

- The distribution of the standardized variable $Z = (Y - u)/b$
  - Probability density function of Z

$$f_Z(z) = f_Y\left(\frac{y-u}{b}\right)\left|\frac{dy}{dz}\right| = (1/b)\, f_0(z)\,(b) = f_0(z)$$

  - Survivor function of Z

$$S_Z(z) = \int_z^\infty f_0(x)\, dx = S_0(z)$$

  - Cumulative density function of Z

$$F_Z(z) = F_0(z)$$

Log-Location-scale Models

- There is an one-to-one correspondence between some lifetime and log-lifetime distributions

Lifetime (T)		log-Lifetime (Y)
Weibull	$\longleftrightarrow$	extreme value
log-logistic	$\longleftrightarrow$	logistic
log-normal	$\longleftrightarrow$	normal

- Parameters of *lifetime* distributions

$$\text{scale } (\alpha) \text{ and shape } (\beta)$$

- Parameters of *log-lifetime* distributions

$$\text{location } (u = \log \alpha) \text{ and scale } (b = 1/\beta)$$

Log-Location-scale Models

- For the standardized log-lifetimes $Z = (Y - u)/b$
- The density, cumulative density, and survivor functions can be expressed in terms of $f_0(\cdot)$, $F_0(\cdot)$, and $S_0(\cdot)$, respectively
- For example, the survivor functions of log-lifetimes are defined as

$$S_0(z) = \exp(-e^z) \longrightarrow \text{ extreme value}$$
$$S_0(z) = 1 - \Phi(z) \longrightarrow \text{ normal}$$
$$S_0(z) = (1 + e^z)^{-1} \longrightarrow \text{ logistic}$$

Log-Location-scale Models

- Using the transformation $T = \exp(Y)$, lifetime distributions can be obtained from each of the distributions of location-scale family

$$
\begin{aligned}
S_T(t) &= P(T \geq t) \\
&= P(\log T \geq \log t) \\
&= S_0\left(\frac{\log t - u}{b}\right) \\
&= S_0^\star\left(\left(\frac{t}{\alpha}\right)^\beta\right)
\end{aligned}
$$

  - $S_0^\star(x) = S_0(\log x)$

Log-Location-scale Models

- Obtain the survivor function of $T \sim \text{Weib}(\alpha, \beta)$ from $Y \sim EV(u, b)$

$$
\begin{aligned}
S(t) &= S_0^{\star}\Big((t/\alpha)^{\beta} \Big) \\
&= S_0\Big(\log (t/\alpha)^{\beta} \Big) \\
&= \exp\Big(- e^{\log \left(t/\alpha\right)^{\beta}} \Big) \\
&= \exp\Big(- (t/\alpha)^{\beta} \Big)
\end{aligned}
$$

- Similarly, obtain the expressions of survivor function of log-logistic and log-normal distribution using the relationship $S(\cdot) = S_0^{\star}(\cdot)$

Subsection 4

1.4 Regression Models

1.4 Regression Models

- Regression models are used to understand the relationship between lifetime and a set of covariates (e.g. age, gender, disease status, values of bio-markers, etc.), some of which may depend on time
- Regression models considered for lifetimes can be divided into two broad categories
  - Parametric models
  - Semiparametric models

Parametric Regression Models

- Parametric models discussed in this chapter (e.g. Weibull, log-logistic, etc.) can be considered for modeling lifetime
- In parametric regression model, one of the parameters of the assumed lifetime distribution is expressed as a function of available covariates

Parametric Regression Models

- Let T be the lifetime and $\mathbf{x} = (x_1, \dots, x_p)'$ be the available p covariates
- Assume $T \sim \text{Exp}(\theta)$ and since $\theta > 0$, a reasonable model for θ would be
$$\theta(\mathbf{x}) = \exp\left(\beta' \mathbf{x}\right), \qquad \text{where } \beta = (\beta_1, \dots, \beta_p)'$$
- The model specification $\theta(\mathbf{x}) = \exp\left(\beta' \mathbf{x}\right)$ ensures $\theta(\mathbf{x}) \geq 0$ for any set of values of β and $\mathbf{x}$
- For the given set of covariates $\mathbf{x}$, the survivor function is defined as
$$S(t \,|\, \mathbf{x}) = \exp(-t/\theta(\mathbf{x}))$$

Parametric Regression Models

- If $Y = \log T$ follows a distribution of location-scale family, the model $u(\mathbf{x}) = \beta^{'}\mathbf{x}$ would be useful, $-\infty < u(\mathbf{x}) < \infty$
- The corresponding survivor function has the form

$$S_Y(y \,|\, \mathbf{x}) = P(Y \geq y \,|\, \mathbf{x}) = S_0\left(\frac{y - u(\mathbf{x})}{b}\right)$$

  - For example, if $S_0(\cdot)$ is the survivor function of standard normal distribution, then the model $u(\mathbf{x}) = \beta^{'}\mathbf{x}$ represents the multiple linear regression model!

Semiparametric Regression Models

- In semiparametric regression model, the dependence of Y or T on $\mathbf{x}$ is specified by a parametric function without making any distributional assumption regarding Y or T
- For lifetime data, the most famous semiparametric regression model is Cox's proportional hazards model (Cox 1972)
- Cox's model considers the hazard function of T given $\mathbf{x}$ of the form

$$h(t \,|\, \mathbf{x}) = h_0(t) \, \exp(\beta' \mathbf{x})$$

  - $h_0(t) \longrightarrow$ arbitrary "baseline" hazard function
  - Time-dependent covariates can be included in Cox's proportional hazards model

Exercises

1. Obtain graphs of probability density, survivor, and cumulative hazard functions of the following distributions using R codes.

a. Weibull distribution with
   i. scale parameter 10, and shape parameter 1.5 and
   ii. scale parameter 10, and shape parameter 0.95
b. Logistic distribution with
   i. location parameter 10 and scale parameter 1.5 and
   ii. location parameter 10 and scale parameter 0.75

Acknowledgements

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